Your grade: 92%

Your latest: **92%** • Your highest: **92%** • To pass you need at least 80%. We keep your highest score.

Next item

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding autoencoders and some aspects of GANs, but there is room for growth in encoding and preprocessing for recommendation systems and understanding GAN challenges. Before retrying this assignment, we recommend focusing on these key areas:

- **Recommender Systems**: In Question 1, your understanding of the common encoding and preprocessing steps for building a movie recommendation system was incomplete. Try reviewing RecSys (Coding 2/4) - Model Training and Evaluation to refresh your memory.
- **Generative Adversarial Networks (GANs)**: In Question 4, your answer about the challenges associated with GANs was incorrect. Try reviewing GANs (101) to better understand the challenges like mode collapse and training instability.

Good job on the rest of the assignment!

Retry assignment

1. What are some common encoding and preprocessing steps required for building a movie recommendation system? — 0.8 / 1 point

- [x] Normalizing ratings

  Correct
  Correct! Normalizing ratings helps in standardizing the data values.

- [ ] Removing stop words
- [ ] Applying collaborative filtering
- [ ] One-hot encoding categorical variables
- [x] Creating embedding layers

  Correct
  Correct! Embedding layers are used to convert high-dimensional categorical data into a lower-dimensional space.

You didn't select all the correct answers

2. When implementing autoencoders for data compression, which of the following represents the bottleneck layer? — 1 / 1 point

- (•) The hidden layer with the smallest number of neurons
- ( ) The input layer
- ( ) The activation layer
- ( ) The output layer

Correct
Correct! The bottleneck layer is the hidden layer with the smallest number of neurons, which forces compression of the input data.

3. What is one primary application of autoencoders? — 1 / 1 point

- ( ) Data encryption
- (•) Dimensionality reduction
- ( ) Generating random numbers
- ( ) Sorting data in ascending order

Correct
Correct! Autoencoders are commonly used for reducing the dimensionality of data.

4. Which of the following are challenges associated with Generative Adversarial Networks (GANs)? — 0.8 / 1 point

- [x] Mode collapse

  Correct
  Correct. Mode collapse is a common issue where the generator produces limited diversity in output.

- [ ] Vanishing gradients
- [ ] Credit assignment
- [x] Training instability

  Correct
  Correct. Training instability is a significant challenge when training GANs.

- [ ] Data labeling

You didn't select all the correct answers

5. In the context of a GAN, what is mode collapse? — 1 / 1 point

- ( ) When the training process becomes unstable and diverges
- ( ) When the discriminator fails to distinguish between real and fake data
- ( ) When the input data is not properly preprocessed
- (•) When the generator produces limited diversity in output

Correct
Correct! Mode collapse occurs when the generator produces outputs that lack diversity, focusing on a limited set of outcomes.